

Theorem scaffold: every architectural choice traces to a named theorem the dissertation measures

Theorems

Takens' embedding theorem

Takens (1981)
A delay reconstruction of
sufficient dimension preserves
the latent attractor's topology.

Maass-Markram separation

Maass, Natschläger, Markram (2002)
A high-dimensional recurrent
network maps distinct inputs
to distinct trajectories.

Oseledec multiplicative ergodic theorem

Oseledec (1968)
Along an ergodic trajectory,
the Lyapunov spectrum is
well-defined.

Echo State Property

Jaeger (2001)
For any input, the reservoir
converges to an input-driven
manifold.

Jaeger memory capacity

Jaeger (2001)
Total recall MC is bounded
above by N ; peaks at an
interior leak rate.

Architectural choices they license

**Use a reservoir, sized to exceed
the latent attractor's Takens dimension.**

Exp B — measured $m^ = 4$, far below $N=256$.*

**Use a high- N , sparse, non-trivial
recurrent matrix.**

*$N=256$, $\sigma=0.05$, density 0.10
(established in Ch. 3/6).*

**Driven λ_1 ('driven'), not autonomous
 $\rho(W)$, is the correct stability object.**

*Exp A — autonomous vs. driven
comparison.*

**$\lambda_1 < 0$ verifies the ESP
under real ERP drive.**

*Exp A — $n=3,165$, median $\lambda_1 = -0.062$,
100% negative.*

**β chosen at the measured MC
plateau, not validation accuracy.**

*Exp C — $\beta=0.05$ within plateau
($0.010 \leq \beta \leq 0.118$, $MC \geq 0.75$).*

*Pesin's identity is deliberately excluded — it licenses claims about systems with positive
Lyapunov exponents, which this dissertation neither measures nor uses (operating regime: $\lambda_1 < 0$).*